

Colby Crutcher

Secure Coding

Lab 5

Task 1

1.

```
ccrutter@cybr437:~/lab5/Task1$ checksec --file=task1
RELRO           STACK CANARY      NX            PIE            RPATH            RUNPATH         Symbols         FORTIFY Fortified   Fo
tifiable       FILE
Full RELRO     No canary found  NX enabled    PIE enabled     No RPATH         No RUNPATH      42 Symbols      No      0              1
task1
```

Figure 1: Checksec

```
ccrutter@cybr437:~/lab5/Task1$ file task1
task1: ELF 64-bit LSB pie executable, x86_64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=3e16b27168ac0fc0513f34b04d3c682b07cd4a92, for GNU/Linux 3.2.0, with debug_info, not stripped
```

Figure 2: File

```
ccrutter@cybr437:~/lab5/Task1$ python3 exploit.py
[+] Starting local process './task1': pid 122721
[+] Receiving all data: Done (63B)
[*] Process './task1' stopped with exit code -11 (SIGSEGV) (pid 122721)

Revealing the secret message:
EWU{overflowing_with_Secrets!}

ccrutter@cybr437:~/lab5/Task1$
```

Figure 3: Successful Output